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COVID-19 Across Pandemic Variant Periods: The Severe Acute Respiratory Infection-Preparedness (SARI-PREP) Study

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IMPORTANCE: The severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) pandemic has evolved through multiple phases in the United States, with significant differences in patient centered outcomes with improvements in hospital strain, medical countermeasures, and overall understanding of the disease. We describe how patient characteristics changed and care progressed over the various pandemic phases; we also emphasize the need for an ongoing clinical network to improve the understanding of known and novel respiratory viral diseases.

OBJECTIVES: To describe how patient characteristics and care evolved across the various COVID-19 pandemic periods in those hospitalized with viral severe acute respiratory infection (SARI).

DESIGN: Severe Acute Respiratory Infection-Preparedness (SARI-PREP) is a Centers for Disease Control and Prevention Foundation-funded, Society of Critical Care Medicine Discovery-housed, longitudinal multicenter cohort study of viral pneumonia. We defined SARI patients as those hospitalized with laboratory-confirmed respiratory viral infection and an acute syndrome of fever, cough, and radiographic infiltrates or hypoxemia. We collected patient-level data including demographic characteristics, comorbidities, acute physiologic measures, serum and respiratory specimens, therapeutics, and outcomes. Outcomes were described across four pandemic variant periods based on a SARS-CoV-2 sequenced subsample: pre-Delta, Delta, Omicron BA.1, and Omicron post-BA.1.

SETTING: Multicenter cohort of adult patients admitted to an acute care ward or ICU from seven hospitals representing diverse geographic regions across the United States.

PARTICIPANTS: Patients with SARI caused by infection with respiratory viruses.

MAIN OUTCOMES AND RESULTS: Eight hundred seventy-four adult patients with SARI were enrolled at seven study hospitals between March 2020 and April 2023. Most patients (780, 89%) had SARS-CoV-2 infection. Across the COVID-19 cohort, median age was 60 years (interquartile range, 48.0–71.0 yr) and 66% were male. Almost half (430, 49%) of the study population belonged to underserved communities. Most patients (76.5%) were admitted to the ICU, 52.5% received mechanical ventilation, and observed hospital mortality was 25.5%. As the pandemic progressed, we observed decreases in ICU utilization (94% to 58%), hospital length of stay (median, 26.0 to 8.5 d), and hospital mortality (32% to 12%), while the number of comorbid conditions increased.

CONCLUSIONS AND RELEVANCE: We describe increasing comorbidities but improved outcomes across pandemic variant periods, in the setting of multiple factors, including evolving care delivery, countermeasures, and viral variants. An understanding of patient-level factors may inform treatment options for subsequent variants and future novel pathogens.

KEYWORDS: clinical research networks; COVID-19; equity; observational study; severe acute respiratory syndrome coronavirus 2

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KEY POINTS

Question: In those admitted with viral severe acute respiratory infection (SARI), how have clinical outcomes and patient characteristics evolved over the various COVID-19 pandemic periods?

Findings: A large, geographically diverse, multicenter, observational study demonstrates that patient-centered outcomes improved over the pandemic periods, even as the prevalence of underlying comorbidities increased. Patients in underserved populations had disproportionate levels of hospital admissions for severe or critical respiratory illness during the pandemic.

Meaning: Improvements in disease understanding, medical countermeasures, surge management, and immunity from prior severe acute respiratory syndrome coronavirus 2 infection or vaccination may have contributed to improved clinical outcomes. The pandemic amplified preexisting disparities in healthcare access. An ongoing clinical network, such as Severe Acute Respiratory Infection-Preparedness (SARI-PREP), can address unanswered questions to inform and improve the clinical management of SARI patients with seasonal influenza, COVID-19, or other respiratory viral diseases.

The COVID-19 pandemic emerged as a defining global health crisis. First reported in late 2019, the spread of the virus was declared a pandemic by the World Health Organization in March 2020, marking the beginning of a new era in infectious diseases with sweeping impacts on health, society, and global economies (1). In the United States, the response to the pandemic has evolved through several stages, with the first pandemic period comprised the ancestral strain of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) until the rollout of targeted vaccinations, with high mortality and stressed healthcare systems (2–5). The Delta variant characterized the second pandemic period due to its significant immune evasion capabilities, which prompted an emphasis on booster vaccinations and reconsideration of critical care protocols (6). The third and fourth pandemic periods were defined by the Omicron variant and its sub lineages. Initially, Omicron BA.1 brought a surge in cases with notable but incomplete immune escape necessitating changes in treatment strategies within critical care settings (7, 8). Subsequent post BA.1 Omicron subvariants further complicated the clinical management of COVID-19 due to their continued evolution to escape neutralization from monoclonal antibody products developed for pre-exposure prophylaxis of SARS-CoV-2 infection or treatment of mild-to-moderate COVID-19 in high-risk outpatients (8). The COVID-19 pandemic is a stark reminder of the critical importance of robust critical care systems, comprehensive public health strategies, and the need for global collaboration in the face of such health emergencies (9).

As part of the Severe Acute Respiratory Infection-Preparedness (SARI-PREP) Study Group, a prospective, longitudinal multicenter cohort study of viral pneumonia, we describe hospitalized ward and ICU patients with laboratory-confirmed viral infection. We evaluate the trends in comorbidities and outcomes throughout different COVID-19 pandemic variant periods, contextualized within the dynamics of changing care delivery, vaccine developments, therapeutic advances, evolving viral variants, and the impact of prior SARS-CoV-2 infections. By delving into patient-level and sequencing data, gathered longitudinally across these pandemic periods, and examining their links with the evolving clinical outcomes, we aim to shed light on potential treatment strategies for future SARS-CoV-2 variants and other novel respiratory viral pathogens associated with severe acute respiratory infection (SARI) requiring hospitalization.

METHODS

Patient Cohort and Study Design

Patients were prospectively enrolled in SARI-PREP, a multicenter, prospective, observational cohort study enrolling hospitalized patients with laboratory-confirmed viral SARI due to SARS-CoV-2, influenza A and B viruses, or other viral respiratory pathogens (ClinicalTrials.gov: NCT04786301), from March 10, 2020, to April 17, 2023, from seven different academic medical centers (10). SARI-PREP inclusion criteria included age older than 18 years, admission to an acute care ward or an ICU, and a clinical syndrome of lower respiratory tract infection defined by the presence of

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fever AND cough AND radiologic infiltrates, accompanied by new OR worsening hypoxia (oxygen saturation < 94% on room air OR new supplemental oxygenation requirement OR increased supplemental oxygenation requirement) (10). All subjects had nasopharyngeal swabs collected for SARS-CoV-2 testing as part of clinical care and a case of COVID-19 was defined as a positive reverse transcription-polymerase chain reaction assay result for SARS-CoV-2. Patients that met the clinical inclusion criteria, clinical suspicion for presumed respiratory viral infection and tested negative for SARS-CoV-2 were classified as non-COVID-19. Patients were enrolled within 72 hours of hospital admission. Nasopharyngeal swabs were also biobanked at the time of study enrollment and SARS-CoV-2 sequencing was completed centrally at the University of Washington Virology laboratory as previously described (11, 12).

Data were collected for patient-level demographic characteristics, which included race, age, and sex. A list of 41 comorbidities were initially collected, but, due to low frequencies (< 5%) for many of the comorbidities, the final list was specified as asthma, chronic kidney disease (CKD), chronic obstructive pulmonary disorder, coronary artery disease, diabetes mellitus, and hypertension. Comorbidities were also coded as any vs. none for the main analyses. Baseline distributions of each variable, including the four pandemic variant periods (based on a SARS-CoV-2 sequenced subsample: pre-Delta, Delta, Omicron BA.1, and Omicron post-BA.1) and the non-COVID-19 population, were displayed as frequencies (percentages) for categorical variables, or as medians (interquartile ranges [IQRs]) for continuous variables, overall and by site.

The research was approved at participating sites by the University of Nebraska Medical Center Institutional Review Board (544-20-FB) in March 27, 2020, for the study titled, Severe Acute Respiratory Infection Preparedness (SARI-PREP). Procedures were followed in accordance with the ethical standards of the responsible committee on human experimentation and with the Helsinki Declaration of 1975. Informed consent was collected on all patients or was waived by the local regulatory board early during the pandemic. The clinical trials registration number is NCT04786301. All procedures were followed in accordance with the ethical standards of the responsible committee on human experimentation and with the Helsinki Declaration of 1975.

Study Outcomes

The prespecified analysis strategy was to evaluate associations between ventilator-free days (VFDs) alive and potential risk factors or interventions using standard models for count data (e.g., Poisson, negative binomial, or zero-inflated models) (10). However, observed VFDs yielded an almost dichotomous distribution with low frequencies for counts between 1 and 27 (34% total across all 27 possible frequencies in that range; **Fig. 1**); over 65% of subjects had either 0 or 28 VFDs (22% and 44%, respectively). We also found some evolution in the distribution of VFDs from a greater distribution early in the pandemic to more binary later in the pandemic (Fig. 1). Ordinal and multinomial logistic models (with different cut-points across the range of VFDs) were subsequently tested through diagnostic plots (to detect outliers and influential points), but none of the preplanned count models (or subsequently tested ordinal or multinomial models) adequately fit the data when treated as counts or as greater than 2 categories. VFDs were therefore coded as any vs. none or 28 VFDs vs. less than 28 VFDs in the final analysis. Therefore, although mortality was not prespecified as the primary outcome (in the published initial analysis plan), it was used as the main outcome for these analyses.

Using data from the Centers for Disease Control and Prevention, we separated COVID-19 patients into four pandemic variant periods based on the predominant SARS-CoV-2 variant, including Pre-Delta (before July 6, 2021), Delta (July 6, 2021, to December 19, 2021), Omicron BA.1 (December 20, 2021, to December 9, 2022), and Omicron post-BA.1 (December 10, 2022, and after). Sequencing of SARS-CoV-2 in nasopharyngeal swabs from 197 patients identified 13 patients with a SARS-CoV-2 variant sequence that was outside the time period of the corresponding predominant variant. These 13 patients were classified into the pandemic variant period that corresponded with the directly sequenced SARS-CoV-2 variant. We completed an analysis testing the associations of four pandemic periods with the outcome of hospital mortality, defined as death in the hospital or discharge to inpatient or outpatient hospice. Secondary outcomes included risk of invasive mechanical ventilation (IMV) on study enrollment and progression to need for IMV during hospitalization (in patients not receiving IMV at study enrollment). The non-COVID-19 patients enrolled throughout the study were combined into one group, irrespective of pandemic periods. While these

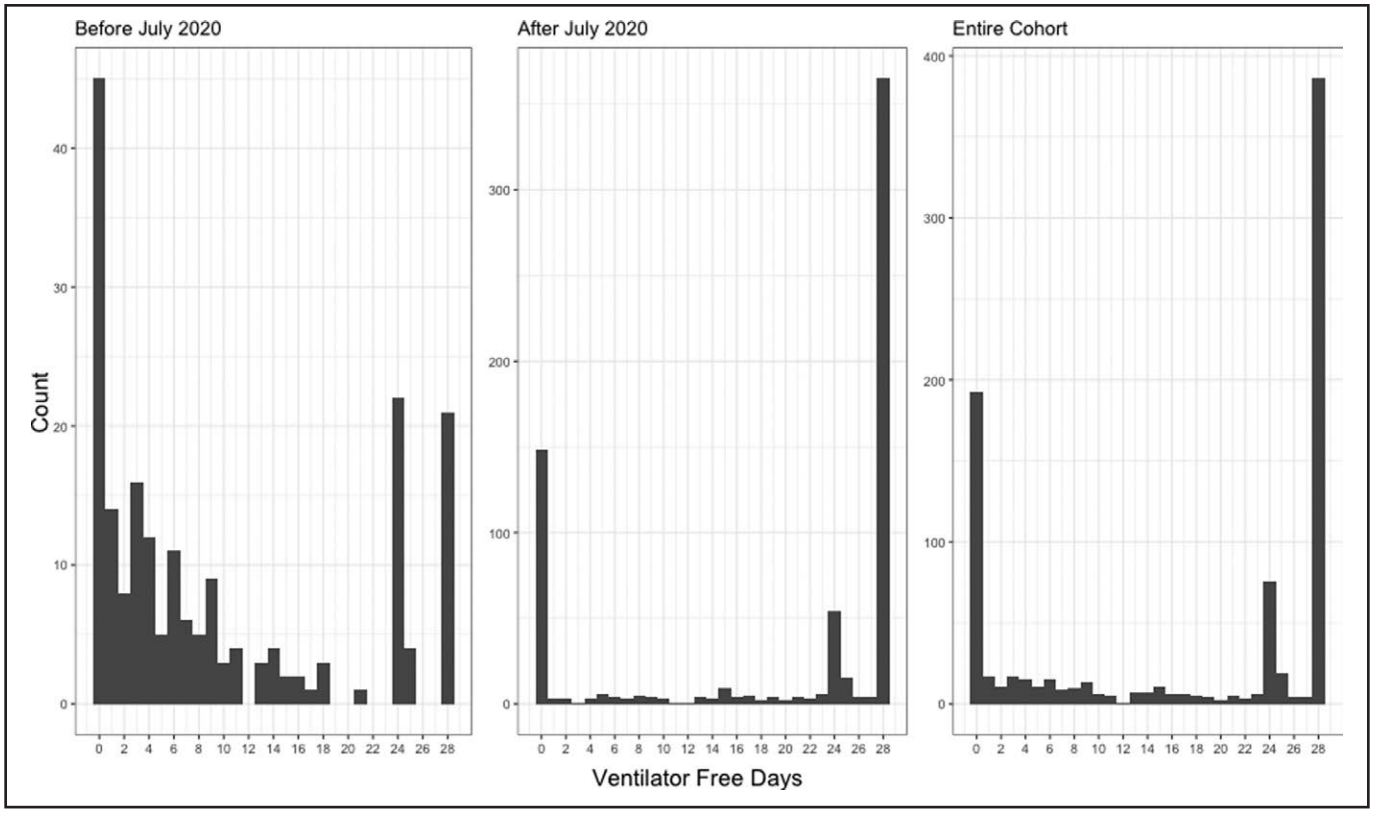


Figure 1. Observed ventilator-free days (VFDs) yielded an almost dichotomous distribution with low frequencies of counts between day 1 and day 27. The distribution of VFDs evolved during the COVID-19 pandemic with initial distribution being more continuous and then later distribution of VFDs being more binary.

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analysis plans varied from the prespecified protocol, these approaches optimized the utility of the collected data.

Statistical Analysis

Logistical regression models were used to quantify associations for all analyses. Model results included the regression coefficient, 95% CIs, and *p* values for both unadjusted and adjusted models for each outcome. The unadjusted models included only the pandemic variant period. Multivariable models were then formed by: 1) forcing in essential variables (which were selected a priori based on clinical significance); these included age, gender, race, ethnicity, COVID-19 vaccination status, and ICU status at admission and 2) using forward selection to evaluate whether any comorbidities added statistically (using $p < 0.1$ as a cutoff) to the adjusted model. In that forward selection, the most significant comorbidity (if $p < 0.1$) was added first to the reduced model and each remaining variable was then individually tested against the new reduced model to identify the next most significant variable, and the process was continued until no other variables could be added. Any variables with *p* value of greater than 0.2 in the resulting multivariable model were then dropped to from the final multivariable model for that outcome.

RESULTS

Study Population and Characteristics

Among 874 SARI patients enrolled, 780 (89%) had reverse transcriptase-quantitative polymerase chain reaction confirmed SARS-CoV-2 infection and 94 (11%) had other respiratory viral infections, including respiratory syncytial virus (RSV) and influenza A and B. Among all patients, the median age was 60 years (IQR, 48–71 yr), 34% were female, 40% identified as non-Hispanic White, 27% as Hispanic, and 13% as non-Hispanic Black. Patients were enrolled within 72 hours of hospital admission with 50% enrolled from ICUs and 50% from non-ICU medical wards. We found a high prevalence of comorbid conditions, including 147 (17%) with CKD, 437 (50%) with hypertension, 295 (34%) with diabetes mellitus, and 135 (15%) with coronary artery disease. Receipt of any primary COVID-19 vaccination series occurred in 22% of patients (**Table 1**).

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TABLE 1.

Admission Characteristics of Hospitalized Patients in Severe Acute Respiratory Infection-Preparedness, Seven Sites, March 2020 to May 2023—Stratified by Severe Acute Respiratory Syndrome Coronavirus 2 Variant Period and Severe Acute Respiratory Syndrome Coronavirus 2 Infection Status

Characteristic	Patients Admitted for SARS-CoV-2 Infection					Patients Admitted for Non-SARS-CoV-2 Respiratory Virus Infection[a] (March 10, 2020, to April 17, 2023)	Total (March 10, 2020, to April 17, 2023)
	Pre-Delta (March 10, 2020, to July 6, 2021)	Delta (July 6, 2021, to December 19, 2021)	Omicron 1 (December 20, 2021, to December 9, 2022)	Omicron 2 (December 10, 2022, to April 17, 2023)	Total, Patients Admitted for SARS-CoV-2 Infection (March 10, 2020, to April 17, 2023)		
Number of patients[b]	492 (100%)	126 (100%)	112 (100%)	50 (100%)	780 (100%)	94 (100%)	874 (100%)
Age	60 (48.0–70.0)	56 (42.3–66.8)	60 (52.0–72.0)	70.5 (54.3–79.0)	60 (48.0–71.0)	64 (49.3–71.8)	60 (48.0–71.0)
Female	144 (29%)	47 (37%)	36 (32%)	25 (50%)	252 (32%)	46 (49%)	298 (34%)
Race and ethnicity							
NH White	165 (34%)	59 (47%)	53 (47%)	26 (52%)	303 (39%)	43 (46%)	346 (40%)
NH Black	47 (10%)	14 (11%)	15 (13%)	11 (22%)	87 (11%)	23 (25%)	110 (13%)
NH Asian	59 (12%)	8 (6%)	15 (13%)	3 (6%)	85 (11%)	1 (1%)	86 (10%)
Hispanic	141 (29%)	35 (28%)	27 (24%)	6 (12%)	209 (27%)	25 (27%)	234 (27%)
Other	59 (12%)	9 (7%)	2 (2%)	4 (8%)	74 (10%)	2 (2%)	76 (9%)
Missing	21 (4%)	1 (1%)	0 (%)	0 (%)	22 (3%)	0 (%)	22 (3%)
Pregnant	3 (1%)	0 (%)	2 (2%)	0 (%)	5 (1%)	1 (1%)	6 (1%)
Current or former smoker (tobacco or vaping)	64 (13%)	42 (33%)	54 (48%)	19 (38%)	179 (23%)	48 (51%)	227 (26%)
SARS-CoV-2 vaccination	13 (3%)	45 (36%)	80 (71%)	26 (52%)	164 (21%)	25 (27%)	189 (22%)
Comorbidities	(%)	(%)	(%)	(%)	(%)	(%)	(%)
Asthma	49 (10%)	12 (10%)	14 (13%)	7 (14%)	82 (11%)	17 (18%)	99 (11%)
Chronic kidney disease	72 (15%)	16 (13%)	30 (27%)	14 (28%)	132 (17%)	15 (16%)	147 (17%)
Chronic obstructive pulmonary disorder	27 (6%)	8 (6%)	9 (8%)	8 (16%)	52 (7%)	14 (15%)	66 (8%)
Coronary artery disease	76 (15%)	9 (7%)	28 (25%)	7 (14%)	120 (15%)	15 (16%)	135 (15%)
Diabetes	181 (37%)	28 (22%)	42 (38%)	17 (34%)	268 (34%)	27 (29%)	295 (34%)
Hypertension	225 (46%)	54 (43%)	71 (63%)	33 (66%)	383 (49%)	54 (57%)	437 (50%)

(*Continued*)

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TABLE 1. (*Continued*)

Admission Characteristics of Hospitalized Patients in Severe Acute Respiratory Infection-Preparedness, Seven Sites, March 2020 to May 2023—Stratified by Severe Acute Respiratory Syndrome Coronavirus 2 Variant Period and Severe Acute Respiratory Syndrome Coronavirus 2 Infection Status

Characteristic	Patients Admitted for SARS-CoV-2 Infection					Patients Admitted for Non-SARS-CoV-2 Respiratory Virus Infection[a] (March 10, 2020, to April 17, 2023)	Total (March 10, 2020, to April 17, 2023)
	Pre-Delta (March 10, 2020, to July 6, 2021)	Delta (July 6, 2021, to December 19, 2021)	Omicron 1 (December 20, 2021, to December 9, 2022)	Omicron 2 (December 10, 2022, to April 17, 2023)	Total, Patients Admitted for SARS-CoV-2 Infection (March 10, 2020, to April 17, 2023)		
Hospital admission location							
Non-ICU medical ward	211 (43%)	71 (56%)	75 (67%)	27 (54%)	384 (49%)	54 (57%)	438 (50%)
ICU	281 (57%)	55 (44%)	37 (33%)	23 (46%)	396 (51%)	40 (43%)	436 (50%)
Acute respiratory distress syndrome at admission	326 (66%)	33 (26%)	12 (11%)	2 (4%)	373 (48%)	2 (2%)	375 (43%)

NH = non-Hispanic, SARS-CoV-2 = severe acute respiratory syndrome coronavirus 2.

[a]Among $n = 55$ with available influenza and respiratory syncytial virus (RSV) testing: $n = 29$ with influenza A, $n = 1$ with influenza B, $n = 7$ with RSV, $n = 1$ with influenza and RSV co-infection, and $n = 17$ without influenza or RSV.

[b]Patients recruited across seven study sites: New York University, University of Washington, University of Southern California, Denver Health, University of Arizona, University of California Irvine, and Emory University.

Values are expressed as *n* (%) or median (interquartile range).

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Four Pandemic Periods and Differences in Patient Characteristics

Seven hundred eighty COVID-19 SARI patients were enrolled across four pandemic variant periods. Patients were classified within a variant period based on date of SARS-CoV-2 diagnosis and a subsample of patients underwent SARS-CoV-2 sequencing of nasopharyngeal swabs (n = 197). The four pandemic variant periods included: 1) pre-Delta (March 10, 2020, to July 5, 2021, n = 492); 2) Delta (July 6, 2021, to December 19, 2021, n = 126); 3) Omicron BA.1 (December 20, 2021, to December 9, 2022, n = 112); and 4) Omicron post-BA.1 December 10, 2022, to April 17, 2023, n = 50) (Table 1).

Compared with the pre-Delta and Delta pandemic periods, patients hospitalized in the Omicron (BA.1 and post-BA.1) periods were older (pre-Delta: 60 ± 22 yr and Delta: 56 ± 25 yr vs. BA.1: 60 ± 20 yr and post-BA.1: 71 ± 25 yr) with a larger proportion of non-Hispanic Black patients (pre-delta 10% and Delta [11%] vs. BA.1 [13%] and BA.2 [22%]). Compared with pre-Delta and Delta, patients hospitalized in the Omicron periods had an overall higher proportion with CKD (15% and 13% vs. 27% and 28%) and chronic obstructive pulmonary disease (6% and 6% vs. 8% and 16%). In comparison, we found that the non-COVID-19 patients had on average a similar demographic characteristics and comorbidities as patients with COVID-19.

Pandemic Periods and Clinical Outcomes

Compared with pre-Delta and Delta periods, patients hospitalized during the Omicron periods had lower or similar ICU utilization during hospitalization (94% and 58% vs. 47% and 58%), shorter hospital length of stay (median, 26.0 d [IQR, 13.0–48.0 d] and 11.0 d [6.0–17.0 d] vs. 9.0 d [5.0–17.0 d] and 8.5 d [5.0–14.0 d]), and lower hospital mortality (32% and 25% vs. 20% and 12%) (**Table 2**). We also found that receipt of IMV at study enrollment (31% and 24% vs. 8% and 4%) and progression to IMV during hospitalization among those not initially intubated (42% and 14% vs. 18% and 8%) was numerically lower in patients hospitalized during the Omicron periods compared with the pre-Delta and Delta periods (Table 2).

Next, we sought to determine if known risk factors for COVID-19-related outcomes, such as demographic characteristics, comorbidities, COVID-19 vaccination status, or severity of illness, could explain the improvement in clinical outcomes during the course of the pandemic. Compared with the pre-Delta pandemic period, we found a stepwise decrease in risk of hospital mortality progressing from the Delta period to post-BA.1 Omicron period (**Table 3**). For example, compared with the pre-Delta period, patients hospitalized during the post-BA.1 Omicron period had a significantly decreased risk of in-hospital mortality (adjusted odds ratio [aOR], 0.18; 95% CI, 0.08–0.38). We also found a consistent improvement in risk of IMV on study enrollment and in progression to IMV during hospitalization across the pandemic periods (**Supplementary Tables 1** and **2**, http://links.lww.com/CCX/B371).

We also compared the risk of clinical outcomes across the pandemic periods to SARI participants with non-SARS-CoV-2 etiologies (non-COVID-19 population) such as influenza A or B virus and RSV infections (**Fig. 2**). The non-COVID-19 patients were enrolled throughout the study (**Supplementary Table 3** and **4**, http://links.lww.com/CCX/B371). Compared with non-COVID-19, COVID-19 patients had a significantly different risk of hospital mortality in unadjusted and adjusted models (**Table 4**). Participants in the pre-Delta pandemic period had six times greater risk of hospital mortality than non-COVID-19 patients (aOR, 6.25; 95% CI, 2.72–17.06). The risk of hospital mortality for COVID-19 and non-COVID-19 SARI patients was not significantly different for those hospitalized in the post-BA.1 Omicron period (aOR, 1.63; 95% CI, 0.48–5.60). Among patients who received IMV, we found a similar trend in decreasing risk across the COVID-19 pandemic periods compared with non-COVID-19 participants (**Supplementary Tables 5** and **6**, http://links.lww.com/CCX/B371).

VFDs coded as a binary measure based on the above-described preliminary modeling results are reported in **Supplementary Tables 7** and **8** (http://links.lww.com/CCX/B371).

DISCUSSION

In this multicenter study spanning four COVID-19 pandemic periods across 3 years, we described clinical outcomes of 780 patients admitted to hospitals with SARI caused by SARS-CoV-2 and 94 patients admitted with other viral causes of SARI. We found that overall outcomes such as in-hospital mortality,

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TABLE 2.

Outcomes of Hospitalized Patients in Severe Acute Respiratory Infection-Preparedness, Seven Sites, March 2020 to May 2023—Stratified by Severe Acute Respiratory Syndrome Coronavirus 2 Variant Period and Severe Acute Respiratory Syndrome Coronavirus 2 Infection Status

Characteristic	Patients Admitted for SARS-CoV-2 Infection					Patients Admitted for Non-SARS-CoV-2 Respiratory Virus Infection[a] (March 10, 2020, to April 17, 2023)	Total (March 10, 2020, to April 17, 2023)
	Pre-Delta (March 10, 2020, to July 6, 2021)	Delta (July 6, 2021, to December 19, 2021)	Omicron 1 (December 20, 2021, to December 9, 2022)	Omicron 2 (December 10, 2022, to April 17, 2023)	Total, Patients Admitted for SARS-CoV-2 Infection (March 10, 2020, to April 17, 2023)		
Number of patients	492 (100%)	126 (100%)	112 (100%)	50 (100%)	780 (100%)	94 (100%)	874 (100%)
Hospital length of stay (d)	26 (13.0–48.0)	11 (6.0–17.0)	9 (5.0–16.0)	8.5 (5.0–14.0)	17 (8.0–40.0)	7 (5.0–12.8)	16 (7.0–36.0)
Hospital mortality	158 (32%)	31 (25%)	22 (20%)	6 (12%)	217 (28%)	6 (6%)	223 (26%)
28-d hospital mortality	105 (21%)	25 (20%)	15 (13%)	6 (12%)	151 (19%)	4 (4%)	155 (18%)
Admitted to ICU during stay	464 (94%)	73 (58%)	53 (47%)	29 (58%)	619 (79%)	50 (53%)	669 (77%)
ICU length of stay (d)	18 (7.0–26.0)	10 (7.0–17.0)	8 (4.0–16.0)	5 (4.0–10.0)	14.5 (6.0–25.0)	7 (4.0–13.0)	14 (5.0–24.0)
ICU mortality	157 (32%)	30 (24%)	19 (17%)	3 (6%)	209 (27%)	4 (4%)	213 (24%)
Oxygenation Methods Used During Stay							
Invasive mechanical ventilation	358 (73%)	48 (38%)	28 (25%)	6 (12%)	440 (56%)	19 (20%)	459 (53%)
Noninvasive mechanical ventilation	23 (5%)	11 (9%)	22 (20%)	5 (10%)	61 (8%)	18 (19%)	79 (9%)
High-flow nasal cannula	99 (20%)	59 (47%)	58 (52%)	14 (28%)	230 (30%)	23 (25%)	253 (29%)
Length of invasive mechanical ventilation use (d)	13 (4.0–23.0)	12 (8.0–20.0)	9.511 (4.8–19.3)	6 (3.5–7.8)	13 (4.0–23.0)	9 (4.0–14.0)	12 (4.0–23.0)
Length of noninvasive mechanical ventilation use (d)	4 (2.0–5.0)	2 (1.0–4.0)	4 (2.3–5.8)	2 (2.0–2.0)	3 (2.0–5.0)	3 (2.0–6.3)	3 (2.0–5.0)
Length of high-flow nasal cannula use (d)	6 (4.0–11.0)	6 (3.0–8.5)	6.5 (3.0–9.5)	3.5 (2.0–6.5)	6 (3.0–10.0)	4 (2.0–8.5)	6 (3.0–10.0)
IMV at admission	152 (31%)	30 (24%)	9 (8%)	2 (4%)	193 (25%)	14 (15%)	207 (24%)
Progression to IMV	207 (42%)	18 (14%)	20 (18%)	4 (8%)	249 (32%)	5 (5%)	254 (29%)
Ventilator-free days, mean (SD)	13.4 (11.9)	19.1 (1220%)	22.3 (10.4)	24.5 (9.2)	16.3 (12.2)	24.9 (7.4)	17.2 (12.1)

IMV = invasive mechanical ventilation, SARS-CoV-2 = severe acute respiratory syndrome coronavirus 2.

[a]Among $n = 55$ with available influenza and respiratory syncytial virus (RSV) testing: $n = 29$ with influenza A, $n = 1$ with influenza B, $n = 7$ with RSV, $n = 1$ with influenza and RSV co-infection, and $n = 17$ without influenza or RSV.

Values are expressed as *n* (%) or median (interquartile range).

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TABLE 3.

Risk of Hospital Mortality by Severe Acute Respiratory Syndrome Coronavirus 2 Variant Period in Severe Acute Respiratory Syndrome Coronavirus 2 Severe Acute Respiratory Infection Patients in Severe Acute Respiratory Infection-Preparedness, Seven Sites, March 2020 to May 2023

	Number of Patients		Unadjusted Model		Adjusted Model[a]	
	n	Event *n*	OR (95% CI)	*p*	OR (95% CI)	*p*
Severe acute respiratory syndrome coronavirus 2 variant period						
Pre-Delta	494	159	Reference		Reference	
Delta	131	31	0.65 (0.41–1.01)	0.061	0.88 (0.53–1.43)	0.604
Omicron 1[b]	153	24	0.39 (0.24–0.62)	< 0.001	0.51 (0.27–0.90)	0.024
Omicron 2[c]	95	8	0.19 (0.08–0.39)	< 0.001	0.18 (0.08–0.38)	< 0.001

OR = odds ratio.

[a]Results of multivariable logistic regression models, adjusted for age, sex, race/ethnicity, primary COVID-19 vaccination series receipt, and ICU status at admission (yes/no).

[b]Omicron BA.1 period.

[c]Omicron period post-BA.1.

Severe acute respiratory infection is defined as an acute respiratory infection with either a history of fever or a measured fever of ≥ 38°C and cough, with onset within the last 10 d and requires hospitalization.

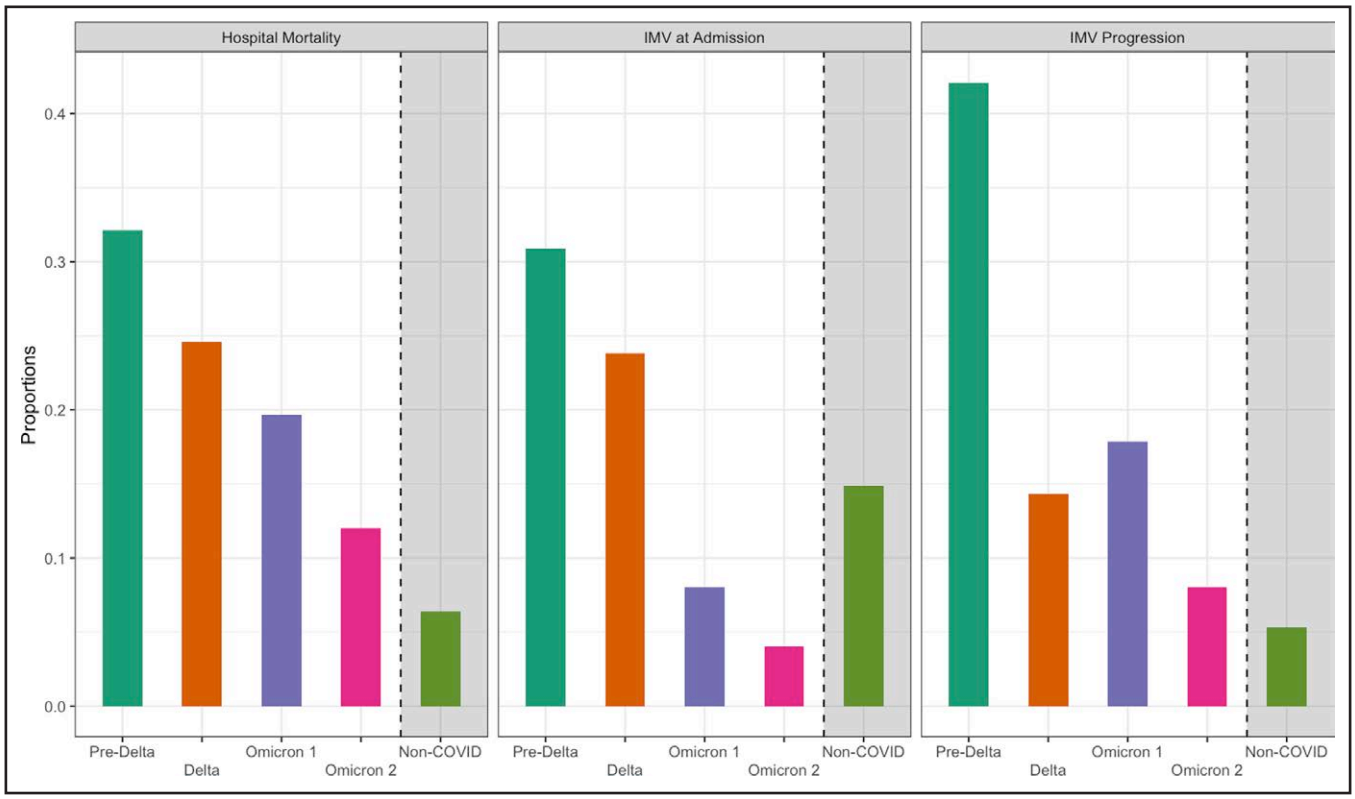


Figure 2. Risk of clinical outcomes across the pandemic periods to severe acute respiratory infection participants with nonsevere acute respiratory syndrome coronavirus 2 etiologies (non-COVID-19 population) such as influenza A or B virus and respiratory syncytial virus infections. IMV = invasive mechanical ventilation.

need for IMV, and progression to IMV after study enrollment all improved over the pandemic periods despite increases in patient ages and prevalence of comorbidities. We also found in multivariable analyses that the improvement in clinical outcomes could not be entirely explained by differences in demographics,

TABLE 4.

Risk of Hospital Mortality Comparing Nonsevere Acute Respiratory Syndrome Coronavirus 2 Severe Acute Respiratory Infection Patients to Severe Acute Respiratory Syndrome Coronavirus 2 Severe Acute Respiratory Infection Patients by Severe Acute Respiratory Syndrome Coronavirus 2 Variant Period in Severe Acute Respiratory Infection-Preparedness, Seven Sites, March 2020 to May 2023

	Counts		Unadjusted Model		Adjusted Model[a]	
	n	Event *n*	OR (95% CI)	*p*	OR (95% CI)	*p*
SARS-CoV-2 variant period						
Non-SARS-CoV-2	94	6	Reference		Reference	
Pre-Delta	492	158	6.94 (3.22–18.11)	< 0.001	6.25 (2.72–17.06)	< 0.001
Delta	126	31	4.79 (2.03–13.22)	< 0.001	5.66 (2.34–15.98)	< 0.001
Omicron 1[b]	111	21	3.42 (1.39–9.69)	0.011	4.04 (1.60–11.69)	0.005
Omicron 2[c]	50	6	2.00 (0.59–6.74)	0.253	1.63 (0.48–5.60)	0.426

OR = odds ratio, SARS-CoV-2 = severe acute respiratory syndrome coronavirus 2.

[a]Results of multivariable logistic regression models, adjusted for age, sex, race/ethnicity, primary COVID-19 vaccination series receipt, and ICU status at admission (yes/no).

[b]Omicron BA.1 period.

[c]Omicron period post-BA.1.

Severe acute respiratory infection is defined as an acute respiratory infection with either a history of fever or a measured fever of ≥ 38°C and cough, with onset within the last 10 d and requires hospitalization.

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comorbidities, primary COVID-19 vaccination series status, or clinical severity of illness at presentation.

Our study confirms earlier published results of decreasing risk of severe outcomes across pandemic periods (13). There are multiple potential reasons to explain the improvement in clinical outcomes observed across COVID-19 pandemic periods. First, medical countermeasures such as antivirals and vaccines were not available during the early pandemic periods. Second, evolution of the SARS-CoV-2 virus from the initial ancestral/alpha variant to the current Omicron variant lineages has been linked to reduced severity of illness (14). Third, the first two pandemic periods placed hospital systems under immense strain, often leading to contingency or crisis standards of care including deviation from critical care best practices and staffing norms (15). Such supply-demand mismatch has immense implications on patient outcomes with nearly one in four COVID-19 deaths potentially attributable to such surges (16). Fourth, general understanding of management of COVID-19-induced acute respiratory distress syndrome has advanced over time and current supportive care to critically ill patients may be superior to the care offered in the early periods of the pandemic. Finally, prior SARS-CoV-2 infection can generate a protective immune response during reinfection, thereby providing some degree of protection against severe disease (17).

Another finding in our study is the continuing healthcare disparities among patients who develop severe or critical COVID-19 and non-COVID-19 patients. Approximately 50% of patients with and without COVID-19 were from racial and ethnic minorities; this is disproportionate to the distribution of race in the U.S. population (18). These findings are consistent with other studies that have highlighted disparities in patient-centered COVID-19 outcomes such as higher ICU admission and mortality rates for some racial and ethnic minorities (19–21).

Our study has multiple strengths. First, recruitment of patients across multiple geographically discrete regions in the United States allows us to describe a widely diverse cohort representing multiple races and ethnicities. Second, we were also able to describe patient-centered clinical outcomes across multiple years of the pandemic because of the longitudinal nature of the study. Third, having a collaborative network such as SARI-PREP that was mobilized rapidly after the discovery of the emergence of a pandemic virus sets a clear precedent for how clinical research

during a public health emergency can be operationalized. Fourth, multiple SARI-PREP sites have collected serial biologic specimens from enrolled patients that will allow further characterization of the pathobiology, immune responses, and biomarkers associated with clinical outcomes of SARS-CoV-2 infection and other respiratory viruses.

Our study has limitations. While we recognize that outcomes improved during the course of the pandemic and hypothesize regarding the various factors behind such improvement, we cannot ascertain which factors contributed the most to our findings. There are many important competing secular trends in the SARS-CoV-2 virus and COVID-19 management not captured in our data that prevent making causal statements regarding variant-specific mortality or models of care over time. Second, enrolled patients may not have represented all hospitalized SARI patients at participating clinical sites, and over the course of the study, we saw a decline in enrollment that may have been due to hesitancy in research or the changing demographics of the pandemic. Third, our study was mostly centered across large academic medical centers that had preexisting infrastructure to participate in clinical research; consequently, community hospitals were disproportionately under-represented. Fourth, a majority of patients were admitted for severe COVID-19 SARI; therefore, our results are not generalizable to other causes of viral SARI. Fifth, while we collected COVID-19 vaccination data for enrolled patients, we were unable to assess pre-enrollment SARS-CoV-2 immune factors such as prior SARS-CoV-2 infection, time since prior infection, receipt of COVID-19 booster vaccinations, and time since last vaccination. Similarly, while we compared outcomes in SARI patients with COVID-19 to those without COVID-19, we recognize that differences in immunity, processes of care, and surge capacity may have important roles to play in this comparison. Last, we do not have data on how individual hospitals handled their respective surge conditions, which may contribute to patient centered outcomes (16).

In summary, we described a large cohort of patients hospitalized for SARI caused by infection with SARS-CoV-2 or other respiratory viruses and report improved outcomes for COVID-19 patients despite increased patient comorbidities across multiple pandemic periods. We also established an infrastructure for rapidly deployable clinical research in the face of a novel virus, thereby setting the stage for such prospective observational clinical research studies during a future outbreak due to an emerging respiratory pathogen. An ongoing clinical network such as SARI-PREP could further answer currently unanswered questions to inform and improve the clinical management of SARI patients with seasonal influenza, COVID-19, or other respiratory viral diseases of public health importance.

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